FEDERAL STANDARD

CARTRIDGE, CO-PLANAR, MAGNETIC, TYPE CPII (COMPACT CASSETTE), AUDIO-VISUAL USE OF

This standard was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

S1. Purpose and scope. This standard specifies the application of compact cassettes for audio-visual and education applications to control audio, slide and filmstrip media; and specifies the technical characteristics of the recorded information.

S2. Application. Specifications and procurement documents of magnetic audio tape recording/reproducing equipment for audio-visual use shall specify that all such equipment shall be compatible with magnetic co-planar cartridge conforming to the requirements of this standard.

S3. Referenced documents. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bid or request for proposal shall apply.

Electronic Industries Association (EIA):

Standard RS-399A- May 1975 - Dimensional Standard Co-planar Magnetic Tape Cartridge, Type CPII (Compact Cassette)

(Application for copies should be addressed to the Electronic Industries Association, Engineering Department, 2001 I Street, NW, Washington, DC 20006.)

International Electrotechnical Commission (IEC):

IEC Recommendation Publication 94-(1968) and Amendment 1 (1971) - Magnetic Tape Recording and Reproducing Systems: Dimensions and Characteristics.

IEC Recommendation Publication 94A-(1972) - Cassette for Commercial Tape Records and Domestic Use, Dimensions and Characteristics.

(Application for copies should be addressed to the Bureau Central de la Commission Electrotechnique Internationale, 1, rue de Varembe, Geneve, Suisse, or to the American National Standards Institute, Inc., (ANSI), 1430 Broadway, New York, NY 10018.)

S4. Definitions.

S4.1 Cue tone. A sinusoidal tone burst recorded on the magnetic tape for the control of peripheral equipment associated with the tape player.

S4.2 Visual synchronization. The control of visual images from such devices as slide and filmstrip projectors so that these images are changed in step with the audio.

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S4.3 Visual advance. The change from one visual image to another initiated by a cue tone.

S4.4 Program stop. The stopping of a program by a cue tone for didactic (theory and practice of learning) or other purposes, the program being restarted at a time determined by the user.

S4.5 Automatic switchoff. The removal of power to an audio-visual device at the end of a program which is initiated by a cue tone. This may also include automatic rewind.

S4.6 Audio comparative. A learning system in which response to pre-recorded information is recorded on a separate track and may be replayed and compared with the pre-recorded information.

S4.7 Breakout lugs. Removable tabs at the rear of the compact cassette used for the protection of recorded material from accidental erasure.

S5. General Standards.

S5.1 Cassette dimensions. The dimensions of the compact cassette shall be in accordance with EIA Standard RS-399A- May 1975.

S5.2 Recording characteristics. The recording characteristics for both program audio and cue signals for audio-visual application specified herein, shall be in accordance with IEC Recommendation 94 with Amendment No. 1.

S5.3 Recording format. The designation, position, and dimensions of the magnetic recording tracks shall be in accordance with the requirements for domestic use as given in IEC Recommendation 94A-(1972).

S5.4 Program audio recording level. Program audio recording level shall be -4dB with reference to 250nWb/m at 315Hz.

S5.5 Cue signal recording level.

S5.5.1 Slides and 35mm filmstrip. Cue signal recording level shall be -6 ± 3dB with reference to 250nWb/m at 315Hz.

Note: If cue signals are recorded on track 4 only, the level should be increased to -3 ± 3dB.

S5.5.2 Super 8mm filmstrip. Cue signal recording level shall be 0dB $\pm$ 3dB with reference to 250nWb/m at 315Hz.

S6. Audio Comparative Applications (See S4.6).

S6.1 Track utilization for audio comparative applications. Tracks 1 and 2 shall be pre-recorded for simultaneous reproduction for one direction of travel as the tape is first unwound from the supply reel. Tracks 1 and 2 may be recorded either monophonically or stereophonically. For monophonic recording, tracks 1 and 2 shall be recorded with identical in-phase information or a single recording of the same total width of the two tracks may be used. For stereophonic recording, track 1 shall be recorded with the left channel and track 2 with the right channel and with head gaps in line. Tracks 3 and 4 shall be non-recorded and be available for future recording by the user (see figure 1).

S6.2 Recorded audio tracks protection. To protect the pre-recorded master tracks (tracks 1 and 2) from accidental erasure, breakout lug as shown in figure 2 shall be removed.

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Note: Use of the breakout lug, as shown in figure 2, complies with existing standards for protection of tracks 1 and 2. Since on audio comparative hardware tracks 3 and 4 are the active recording tracks, the breakout lug sensor will have to be located on the opposite side of the cassette as compared with hardware used for making recordings on tracks 1 and 2.

S7. Audio-visual Separate-Track-Synchronization Application.

S7.1 Description. The audio-visual separate-track-synchronization format places the control signals on a separate track from the audio, using the cassette for a single direction of play, providing for the use of a greater number of control functions and allowing for correction or modification of audio or control signals without affecting the other. The audio program information is recorded on one track and cue tones for control of visual advance and other required functions are recorded on a second track. The compact cassette is recorded for playback in one direction of tape travel only.

S7.2 Audio-visual separate-track-synchronization track utilization. Tracks 1 and 2 shall be recorded with audio program information for simultaneous reproduction. The recording may be monophonic or stereophonic with track 1 carrying the left channel and track 2 the right channel. Tracks 3 and 4 shall be identically and simultaneously recorded with cue tones for control of visual advance and other functions. Recording of track 4 alone is permitted (see figure 3).

S7.3 Recorded audio and synchronizing signals protection. Both breakout lugs at the rear of the compact cassette shall be removed to prevent accidental erasure of the recording. The relationship between the sides of the compact cassette and the breakout lugs is as shown in figure 2.

S7.4 Recording data.

S7.4.1 Tape speed. The tape speed shall be 4.76 centimeters per second (1.875 inches per second).

S7.4.2 Cue frequencies. Four cue tone frequencies are assigned: 150 hertz (Hz), 400 Hz, 1000 Hz, and 2300 Hz, with recorded frequency tolerances of $\pm$ 5 percent. The 150 Hz and 1000 Hz frequencies shall be used only for the primary control purposes as defined in this standard. The 400 Hz and 2300 Hz frequencies may be used for secondary control purposes. Total harmonic distortion of cue tone shall not exceed 10 percent when measured without equalization.

S7.4.3 Cue frequency utilization.

a. 150 Hz: Program stop.

b. 1000 Hz: Advance of visual media, each cue advances the visual media one frame or slide.

c. 400 Hz: Secondary frequency; for other assigned functions.

d. 2300 Hz: Secondary frequency; for other assigned functions.

S7.4.4 Cue width.

a. Slide - (see figure 4).

Advance (1000 Hz): 0.45 $\pm$0.07 seconds.
Program stop (150 Hz): 0.45 $\pm$0.07 seconds.

b. 35mm filmstrip - (see figure 4).

Frame advance (1000 Hz): 0.45 $\pm$.07 seconds.
Program stop (150 Hz): 0.45 $\pm$0.07 seconds.

c. Super 8mm filmstrip - (see figure 5).

Frame advance (1000 Hz): 0.0325 seconds $\pm$5 percent.
Program stop (150 Hz): 0.050 seconds $\pm$5 percent.

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S7.4.5 Cue spacing.

a. Between the start of two adjacent advance cues:

Slide (see figure 4) - 1.5 seconds minimum.
35mm filmstrip (see figure 4) - 0.78 seconds minimum.
Super 8mm filmstrip (see figure 5) - 0.0417 seconds minimum.

b. Between end of a stop cue and start of the next advance or stop cue or restart of audio: 2.0 seconds minimum of tape time (i.e. 9.52 centimeters or 3.75 inches of tape). (See figures 4 and 5).

S8. Audio Record-Playback Application. In this application the audio program information is recorded either monophonically or stereophonically for playback at a later time. This application includes both the capability to erase and the prevention of erasure of recorded information.

S8.1 Audio record-playback track utilization (see figure 6). Tracks 1 and 2 shall be used to record audio program information for simultaneous reproduction for one direction of travel as the tape is first unwound from the supply reel. Tracks 3 and 4 shall be used to record audio program information for simultaneous reproduction for the direction opposite from that of tracks 1 and 2. The combination of tracks 1 and 2 or 3 and 4 may be recorded either monophonically or stereophonically. For monophonic recording tracks 1 and 2 or 3 and 4 shall be recorded with identical in-phase information. Stereophonic track utilization shall be as follows:

Tracks 1 and 2 combination:

Track 1 - left channel.
Track 2 - right channel.

Tracks 3 and 4 combination:

Track 3 - right channel.
Track 4 - left channel.

Note: For monophonic recordings, the space between tracks 1 and 2 or 3 and 4 may be omitted, i.e. a single recording or reproducing head of the same total width of the two tracks may be used.

S8.2 Recording protection. It is possible to protect recorded information from accidental erasure by removing appropriate lugs as shown in figure 2. Once the lugs are removed it shall not be possible to erase or record over existing information.

S9. Changes. When a Federal agency considers that this standard does not provide for its essential needs, written request for changing or adding to the standard, supported by adequate justification explaining wherein the standard does not provide for essential needs, shall be sent in duplicate to the General Services Administration, Federal Supply Service, Office of Standards and Quality Control, Washington, DC 20406. The General Services Administration will determine the appropriate action to be taken and will notify the agency.

S10. Conflict with referenced documents. Where the requirements stated in this standard conflict with any requirement in a referenced document, the requirements of this standard shall apply. Nature of conflict between the standard and the referenced document shall be submitted in duplicate to the General Services Administration, Federal Supply Service, Office of Standards and Quality Control, Washington, DC 20406.

S11. Supersession data. This standard supersedes Military Standard MIL-STD-1853A, 9 May 1975.

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Navy - EC
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User Activities:

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Preparing Activity:

GSA-FSS

Civil Agency Coordinating Activities:

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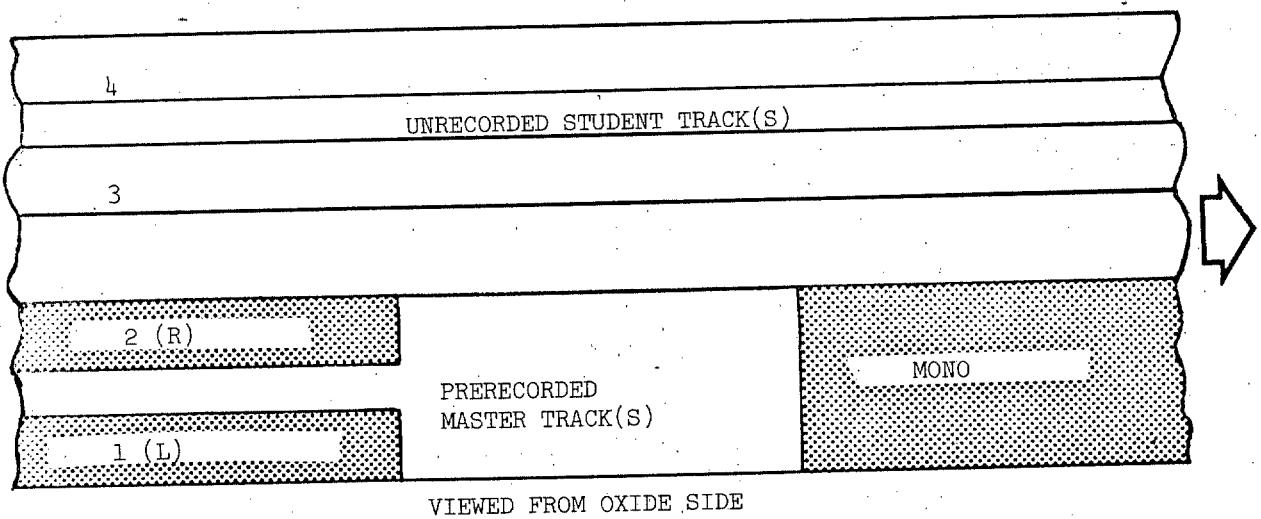
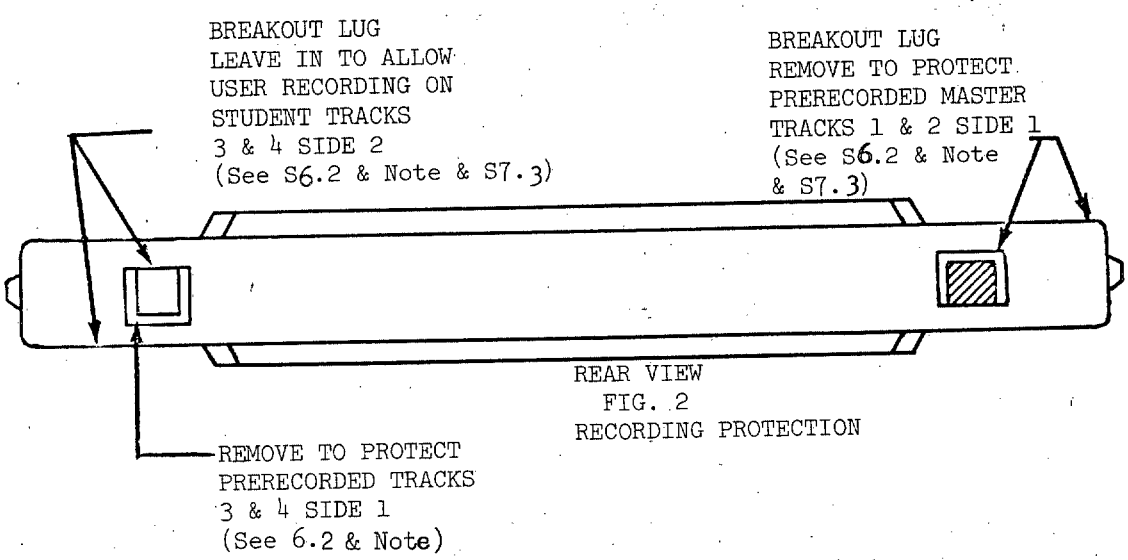
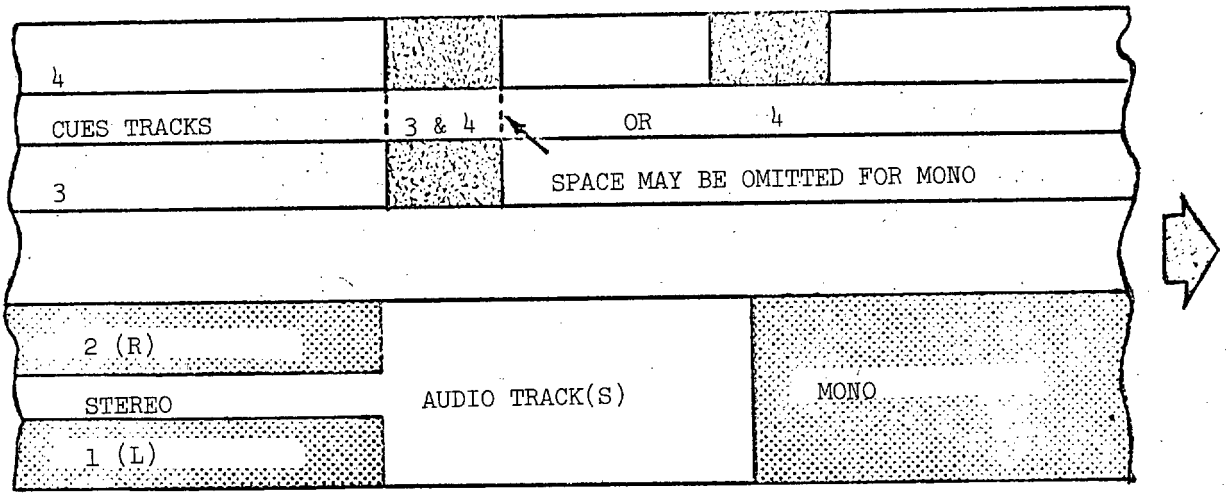


FIG. 1
TRACK UTILIZATION
AUDIO-COMPARATIVE

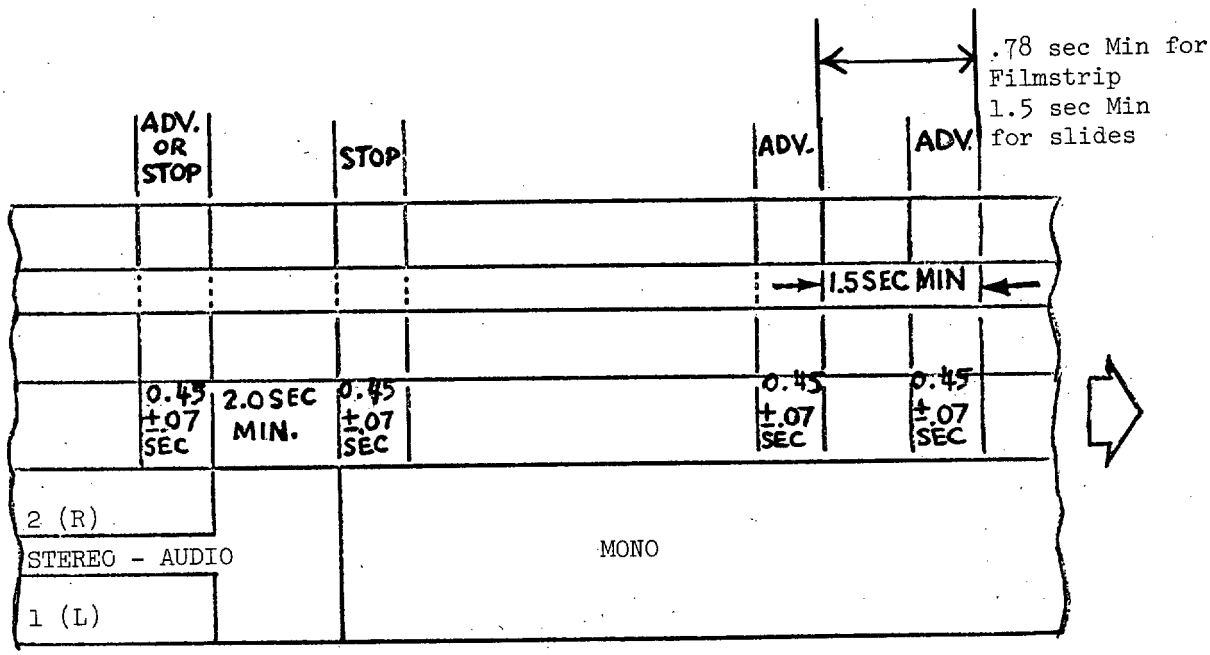


REAR VIEW
FIG. 2
RECORDING PROTECTION

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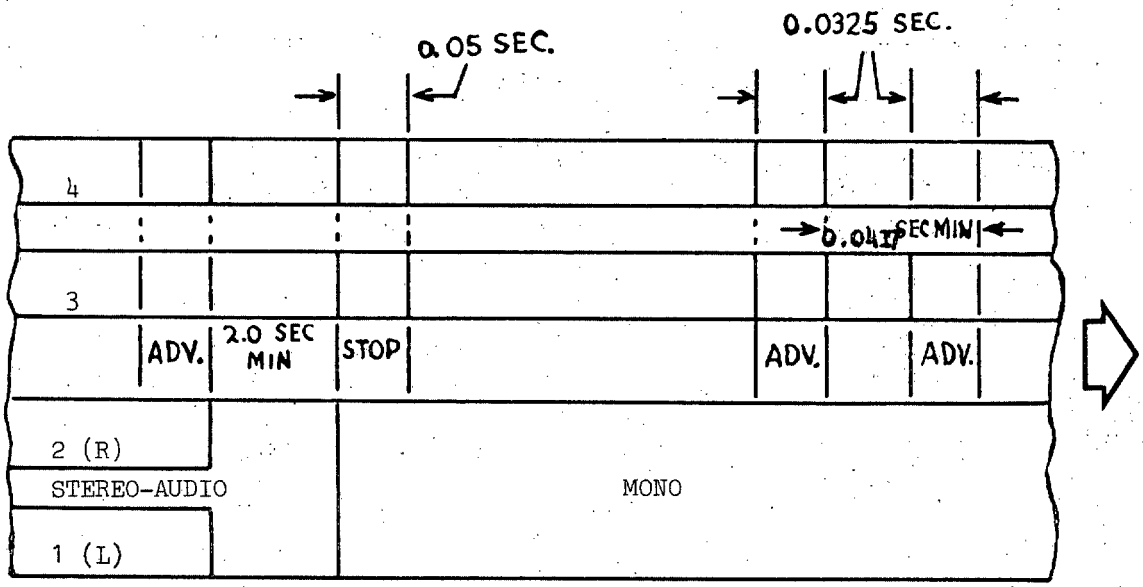


VIEWED FROM OXIDE SIDE
FIGURE 3
TRACK UTILIZATION
AUDIO + VISUAL



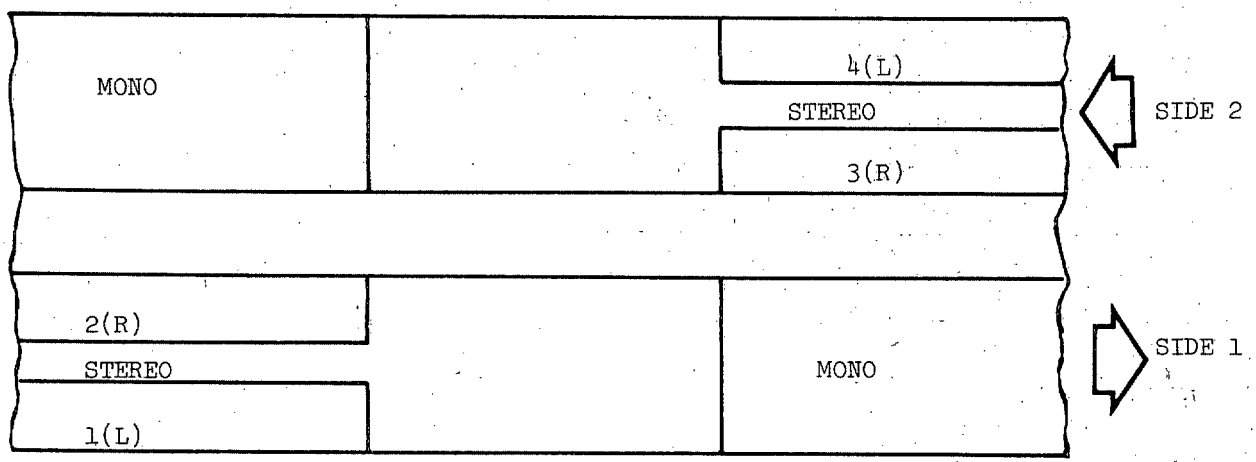
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FIG. 4
CUE WIDTH AND SPACING
SEPARATE TRACK SYSTEM
35MM FILMSTRIP AND SLIDE

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FIG 5
CUE WIDTH AND SPACING,
SEPARATE TRACK SYSTEM
SUPER 8MM FILMSTRIP



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FIG 6
TRACK UTILIZATION
AUDIO RECORD/PLAYBACK

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